

Technology Stack

Date	02 July 2026
Team ID	XXXXXX
Project Name	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5,CSS3	Provides a simple, responsive, and user-friendly interface for entering HDI input values and displaying prediction results.

architecture and provide a brief justification for your choice.

Technology Stack Details

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles user requests, processes input, loads the trained ML model, and returns prediction results efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Joblib (.pkl Model File)	CSV stores the training data, while the trained machine learning model is stored as a .pkl file for prediction.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render	Easy deployment of the Flask application with reliable cloud hosting.

5	Version Control & CI/CD	Git , GitHub	Tracks code changes, supports collaboration, and maintains version history.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Pandas, NumPy, Scikit-learn, Joblib	Pandas and NumPy handle data processing, Scikit-learn trains the ML model, and Joblib saves/loads the trained model.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		